

pH curves of strong and weak monoprotic acids and bases (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to interpret the general shapes of pH curves for all four combinations of strong and weak acids and bases.

In [section R3.1.8 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/), you learned that titration is one of the most widely used techniques in quantitative analysis to determine concentrations of acids and bases. It is used in the food, pharmaceutical and cosmetic industries as an important analytical technique. Titration curves can give a lot of information regarding the acid or base. But do titration curves for all combinations of strong and weak acids and bases share the same characteristics?



Figure 1. Titration is used in the food, pharmaceutical and cosmetic industries.

Credit: Nico Tekol, Getty Images

The curve of a strong acid–strong base titration was examined in [section R3.1.8](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/>). In this section we will look at the curves of titrations of strong acids–weak bases, weak acids–strong bases, and weak acids–weak bases.

The pH curve for a weak acid—strong base titration

Figure 2 shows the pH curve for the addition of a strong base to a weak acid. Can you think of similarities and differences with the titration curve of a strong acid–strong base titration?

The curve starts at pH 3, which is the pH of the weak acid used to produce this pH curve. At first, the addition of the strong base results in the pH staying fairly constant; this is known as the buffer region. Then there is a sharp increase in pH from 7 to 11 and the curve flattens out at pH 13, the pH of a strong base. While the pH at the equivalence point for the titration of a strong acid with a strong base was 7, here it is not: the pH at the equivalence point is greater than 7. This is because of the basic salt that is produced from the neutralisation reaction, for example here, potassium ethanoate, CH_3COOK .

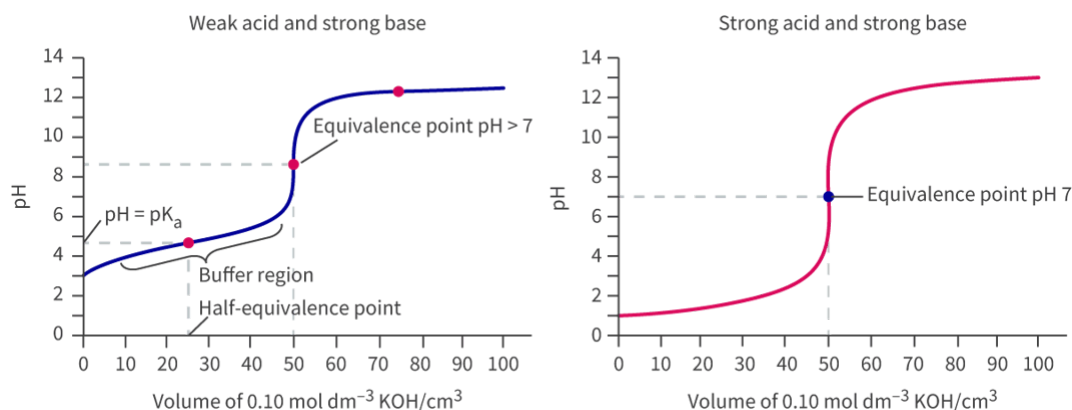


Figure 2. Titration curves for a weak acid and a strong base and strong acid and a strong base.

 More information for figure 2

This image contains two titration curves, each depicting a different chemical reaction.

- Left Graph (Weak Acid and Strong Base):**
- X-axis:** Represents the volume of $0.10 \text{ mol dm}^{-3} \text{ KOH/cm}^3$, ranging from 0 to 100 cm^3 .
- Y-axis:** Represents pH, ranging from 0 to 14.

4. **Curve Description:** Initially starts at a pH of around 3. As KOH is added, the pH remains relatively constant due to the buffer region, indicated between approximately 20 cm³ and 40 cm³. This region is followed by a sharp increase in pH between 40 cm³ and 50 cm³, indicating the equivalence point with a pH greater than 7. The curve flattens at a pH of about 13.
5. **Special Labels:** Marks include "pH = pK_a," "Buffer region," "Half-equivalence point," and "Equivalence point pH > 7."
6. **Right Graph (Strong Acid and Strong Base):**
7. **X-axis:** Similar setup as the first graph with volume of KOH/cm³ from 0 to 100 cm³.
8. **Y-axis:** Similar scale as the first graph with pH from 0 to 14.
9. **Curve Description:** Starts at an acidic pH around 1, increases more significantly after 40 cm³, with the equivalence point at 50 cm³ marked by a pH of 7. Beyond this point, the graph shows a less steep increase, leveling off at a pH near 13.
10. **Special Labels:** Includes "Equivalence point pH 7."

The graphs illustrate the differing behaviors of weak and strong acids when titrated with strong bases, highlighting the buffer region and the varying pH at the equivalence point.

[Generated by AI]

As shown on the first graph of **Figure 2**, there is a region in which pH remains relatively constant labelled as the buffer region. The buffer region is where the solution contains HA (the weak acid) and a significant amount of A⁻ (the conjugate base of the weak acid). At this point, the solution acts as a buffer solution, meaning that it has the ability to resist a change in pH on the addition of a small amount of acid or base. We call this the half-equivalence point.

The half-equivalence point can be found on the x-axis at the point which is half the volume added at the equivalence point. On the pH curve for a weak acid and a strong base in **Figure 2**, this corresponds to the 25 cm³ point. At the half-equivalence point, exactly half the acid has reacted with the base and been neutralised, with the other half remaining unreacted. The weak acid is in excess by half the amount added, the strong base has reacted completely. The conjugate base of the weak acid has been produced from the neutralisation in exactly the same amount as the reacted weak acid. The weak acid and conjugate base are found in solution in equal amounts. Therefore, the properties of the solution at this point make it an ideal buffer solution, which explains why the pH remains fairly constant when a small amount of base is added to the acid.

At the half-equivalence point, the pH is equal to the pK_a of the acid. The pH at the half-equivalence point can be read directly from the graph (draw a line up to the curve, then across to the y-axis), and this can be used to deduce the pK_a of the acid. At this point, the K_a = [H⁺], therefore, the pK_a = pH.

Study skills

When presented with a pH curve, make sure that you are able to find the pH at the equivalence point.

Practical skills

Tool 1: Experimental techniques — Applying techniques

When collecting data to generate a pH curve, when should smaller volumes of titrant be added between each measurement?

pH curve for a weak base—strong acid titration

Based on what you have learned so far, can you try plotting the titration curve for the titration for adding a strong acid to a weak base, for example hydrochloric acid and ammonia?

What do you think the most important features are? Consider the following prompts:

- What do you think the initial pH would be before any acid is added?
- Do you think pH will increase or decrease as more acid is added?
- What do you think the final pH will be after an excess amount of acid is added?
- What do you think the pH at the equivalence point will be, 7, above 7 or below 7?

Use **Interactive 1** to investigate the titration curve for a weak base/strong acid titration.

Interactive 1. The titration curve for a weak base—strong acid titration.

The curve starts at pH 11–12, the pH of the weak base used to produce this pH curve, and then decreases gradually in the buffer region. In this part of the graph, the solution resists changes in pH as the strong acid is added. There is then a steep drop in pH when 25 cm³ of acid has been added. The equivalence point is at a pH of less than 7, because of the acidic salt (NH₄Cl) that is produced.

In the buffer region, at the half-equivalence point, the pH of the conjugate acid of the weak base is equal to pK_a , or $pOH = pK_b$ for the weak base.

Study skills

You should be able to interpret pH curves for all combinations. The interpretation should include:

- intercept with the pH axis
- equivalence point
- buffer region
- points where $pH = pK_a$ or $pOH = pK_b$

pH curve for a weak base—weak acid titration

Based on what you have learned so far, can you try plotting the titration curve for adding a weak acid to a weak base? For example, ethanoic acid and ammonia.

What do you think the most important features are? What will be different from all graphs seen previously?

Use **Interactive 2** to investigate the titration curve for a weak base–weak acid titration.

Interactive 2. The Titration Curve for a Weak Base—Weak Acid Titration.

Interactive 2 shows the pH curve for the addition of a weak acid to a weak base. The curve starts at around pH 12, the pH of the weak base used to produce this pH curve, and then decreases to the pH of the added weak acid. The equivalence point is at a pH of 7, because of the neutral salt produced.

The starting pH is not high, the ending pH is not low, and there is a very small change in pH. The equivalence point is, therefore, difficult to determine and the graph is not particularly useful.

Study skills

It is possible to carry out these titrations the other way round — with the alkali in the flask, and adding the acid from the burette, or with the acid in the flask and adding the alkali from the burette. The curves obtained in this case are simply reflections of those we have considered here.

Practical skills

Tool 2: Technology — Applying technology to collect data

Tool 3: Mathematics - Graphing, Sketch graphs, with labelled but unscaled axis to qualitatively describe trends, Construct and interpret graphs for raw and processed data, Draw lines or curves of

best fit

In the lab, you can perform a titration of a weak acid with a strong base to determine the pK_a of the weak acid experimentally. You can collect pH using a data-logger, then graph the data and draw lines of best fit. How can you use the graph to determine the pK_a ?

Use **Interactive 3** to review the features of different titration curves.

Specify if the titration curve is for a weak acid–weak base, strong acid–strong base, weak acid–strong base, or strong acid–weak base combination.

pH at the equivalence point is 8.5.

Turn

Card 1 of 4



Interactive 3. Review the features of different titration curves.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
 - Thinking skills — Being curious about the natural world; Applying key ideas and facts in new contexts
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes

- **Activity type:** Pair activity

Instructions

1. Open the titration curve generator  (https://iwant2study.org/lookangejss/chemistry/2025_04_14/) which has titrations for six combinations: strong acid—strong base, strong base—strong acid, weak acid—strong base, weak base—strong acid, strong base—weak base, and strong acid—weak base.
2. Set-up each titration by choosing the concentration of acid and base. You can also choose either monoprotic or diprotic and the volume of solution in the conical flask.
3. Choose an indicator that you think will cover the pH range containing the equivalence point.
4. Run each titration and study the pH curves produced.
5. Try each of the different combinations of acids and bases and compare the pH curves produced.